Thermal Demagnetization Studies on the Old Red Sandstone of the Anglo-Welsh Cuvette

F. H. CHAMALAUN[1] AND K. M. CREER

Department of Physics, University of Newcastle-upon-Tyne Great Britain

Abstract. Thermal demagnetization of samples from folded and flat-lying sites of the Old Red Sandstone of the Anglo-Welsh cuvette shows that the natural remanence is composite. There is a weak component possessing high thermal stability, which can be shown to be of pre-Hercynian age by application of the fold test and which can only be detected by heating to above 600°C. There is also a second and dominant component thermally less stable, which is almost entirely responsible for the natural remanence and which represents a secondary magnetization imposed on the rocks at some later date. The direction of the more stable component agrees with the direction of the natural remanent magnetization of the Lower Old Red Sandstone lavas from the Midland Valley of Scotland. It is therefore suggested that the paleomagnetic pole position (140°E, 10°S) obtained from the Lower Old Red Sandstone lavas is the best estimate of the Devonian paleomagnetic pole. That previously suggested at 159°E, 30°N, which was deduced from the remanent magnetization of the Old Red Sandstones of the Anglo-Welsh cuvette, is shown to be appropriate to a later paleomagnetic field. The discrepancy between the revised Devonian pole for Britain and Devonian data from other parts of Europe, notably Russia, is noted.

Introduction. Red beds have been extensively used in paleomagnetic research. However, the age of their natural remanent magnetization (NRM) is often unknown because it is possible that this might be a chemical remanent magnetization and therefore younger than the geological age of the rocks. This is in contrast with igneous rocks which usually acquire their magnetization on cooling through the Curie point of the magnetic minerals contained in them. Two methods exist for testing the age of the NRM of sedimentary rocks. One is based on field tests, such as the fold test and conglomerate test [*Graham,* 1949], and the other is based on the comparison between igneous rocks and sedimentary rocks of the same age. In this respect the Devonian of Great Britain poses several problems. First, the fold test applied to the folded Old Red Sandstone strata in Pembrokeshire failed to indicate whether the magnetization was acquired before or after folding. Second, the direction of magnetization of the Lower Devonian lavas from the Midland Valley of Scotland was significantly different from that of the flat-lying Old Red Sandstone beds of the Anglo-Welsh cuvette.

In this paper we describe the results obtained from thermal demagnetization experiments on the latter formation (which were undertaken with a view to clarify this position). Although thermal demagnetization methods have been used by several workers in paleomagnetism [*Thellier and Thellier,* 1959; *Leng,* 1955; *Doell,* 1956; *Irving et al.,* 1961], applications to typical red beds are rare. Also, in previous work the method was applied mainly to rocks of which the distribution of the directions of magnetization strongly suggested partial instability (see for example *Irving et al.* [1961]). In the present paper we illustrate the use of the method in a case in which the secondary magnetization has completely masked the original magnetization.

Geology. The Anglo-Welsh cuvette is roughly triangular. It extends from Milford Haven in the west to the River Severn in the east, and to Bridgenorth in the north. In this region the Devonian consists of the continental facies, the Old Red Sandstone (red beds), sandstones, marls, breccias, and mudstones. These represent the estuarian and deltaic deposits of a river system draining a metamorphic complex to the northwest [*Heard and Davies,* 1924; *Jones,* 1955]. The region has been affected by the Hercynian orogeny (Permo-Carboniferous in

[1] Now at the Department of Geophysics, Australian National University, Canberra, Australia.

age) which attained its greatest intensity in this region to the southwest in Pembrokeshire. As a result, the strata there are steeply folded, dipping vertically, and decreasing in age northward at Manorbier Bay and southward at Freshwater-West. To the northeast the beds are more or less flat lying, the geological tilt being less than 20°.

The samples to be discussed in this paper are of Lower Devonian age with the exception of those from Portishead, which are Upper Devonian. Only hard, fine-grained sandstones were used in the thermal experiments since it was found that the other types tended to disintegrate during heating.

Previous paleomagnetic work on the Devonian in Britain. In a previous survey of the Old Red Sandstone of the Anglo-Welsh cuvette [*Creer*, 1957] it was found that both limbs of the steep anticlinal fold in Pembrokeshire had a direction of magnetization significantly different from each other and from the directions of magnetization of the flat-lying beds. The directions of magnetization are summarized in Table 1.

To explain the differences in the directions between these three sites, it was suggested that the folded strata contained two components of magnetization [*Creer*, 1957]: a primary component which was supposed to have been acquired before folding and to have a direction similar to that found for the flat-lying sites, and a secondary component supposed to have been superimposed on this after folding in a nearly horizontal field. Figure 1 shows how these two combine to give the observed directions. Assuming this *two-component hypothesis*, *Creer* [1962] found the field responsible for the secondary magnetization. From his statistical analysis it was concluded that the geometry of the vectors was such that remagnetization occurred most probably during the Permian or Triassic. It was thought that the secondary magnetization might be a partial thermo-remanent magnetization induced by burial temperatures which could possibly have been as high as 200°C [*Creer*, 1957], and it was decided to attempt to remove this secondary magnetization by means of thermal demagnetization.

TABLE 1. Summary of Paleomagnetic Data for Old Red Sandstone Rocks of the Anglo-Welsh Cuvette

	Declination	Inclination	α_m	No. of Sites
Flat lying	196°	−4°	5°	35
Dipping to the north	210°	+29°	6°	21
Dipping to the south	211°	−32°	8°	14

Note. Declinations are given in degrees east from geographic north α_m radius of confidence at the 95 per cent probability level [*Fisher*, 1953]. The directions are given with respect to the bedding planes.

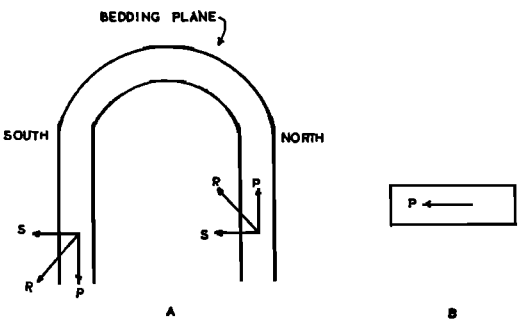


Fig. 1. Schematic representation of the two-component hypothesis. *A*, folded site; *B*, flat-lying site; *P*, primary component parallel to bedding plane; *S*, secondary component; *R*, resultant magnetization.

The thermal demagnetization experiments did not, however, reveal the presence of the two components postulated in the above hypothesis. Instead they showed that some specimens contained a weak but thermally stable component which could be detected only after heating to temperatures as high as 600°C. This component turned out to be relevant to another problem associated with the Devonian paleomagnetism of Britain.

Assuming that the flat-lying beds had not been affected by the Hercynian orogeny and noting that the NRM of the flat-lying sites appeared to possess a high degree of stability, we computed a Devonian paleomagnetic pole from them. This pole, which will be called pole A, was situated at 159°E, 30°N. Other Devonian data from elsewhere in Europe [*Nairn*, 1960] and particularly from Russia [*Kalashnikov*, 1961] were in excellent agreement with pole A, which has therefore generally been accepted as the Devonian pole for Europe.

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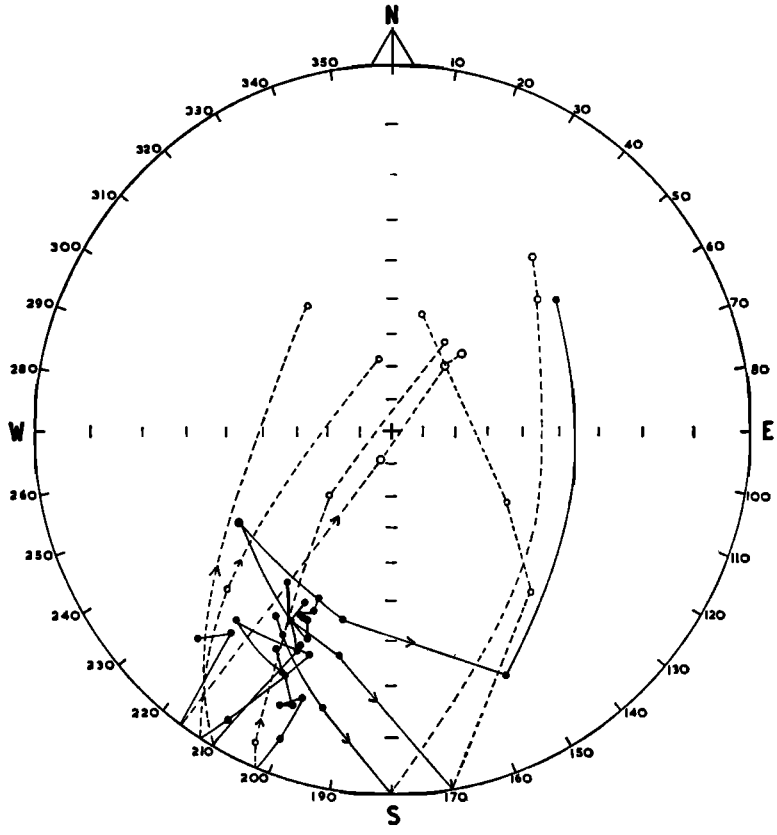


Fig. 2. Change in direction for specimens from sample M5 (Manorbier Bay). In all stereograms the directions are plotted on a stereographic net and with respect to the bedding planes. Closed circles indicate downward (positive) magnetizations and open circles indicate upward (negative) magnetizations.

However a survey by *Stubbs* [1958] on the Lower Old Red Sandstone lavas from the Midland Valley of Scotland yielded a direction of magnetization significantly different from that of the Old Red Sandstones of the flat-lying sites. The corresponding paleomagnetic pole, which will be referred to as pole B, was situated at 140°E, 10°S. Stubbs' investigation comprised 44 flows. The results of 14 flows were rejected on the basis of obvious partial instability. Of the 30 remaining flows, the 16 with least scatter were selected for the calculation of pole B. Eight were normally magnetized and 8 reversely magnetized. These 16 flows cover a thickness of up to 3000 m. There are good reasons for believing that the direction of magnetization is reliable: (1) for lavas there is strong evidence that the magnetic age is the same as the geological age of the rocks; (2) the occurrence of normal and reversed groups whose mean directions are almost exactly at 180° indicates the absence of secondary magnetization. The discrepancy between these two Devonian poles A and B, both of which are for different reasons apparently reliable, has puzzled paleomagnetic workers for many years. It will be shown that the thermal experiments indicate that the weak but stable component of the Old Red Sandstone beds agrees with Stubbs' results, and hence we now attribute the discrepancy to an almost complete remagnetization of the Old Red Sandstones of the Anglo-Welsh cuvette.

Experimental method. The specimens were heated in 8 steps to successively higher temperatures, from 100° to 700°C. After each heating step they were cooled to room temperature

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in zero field and their directions and intensities of magnetization remeasured in the usual way [*Collinson et al.*, 1957]. The earth's field was canceled by means of square-shaped 'Helmholtz' coils [*Parry*, 1957]. Before each heating the earth's field was reduced to less than 10 γ and during the cooling cycle the currents in the coils were monitored and adjusted so that fluctuations in the currents did not cause the field to vary by more than 2 γ from the initial setting. The field was remeasured after each cooling to check that the diurnal variation had not been unduly large.

It was found that heating in nitrogen caused the red cement to be reduced probably to magnetite [*Leng*, 1955; *Chamalaun*, 1963], and therefore the heatings were carried out in air. The furnace, although developed independently, was similar to that used by *Irving et al.* [1961] and was constructed of nonmagnetic materials as far as possible. The heating jacket could be lifted about 1.5 m above the specimens; thus any magnetic parts of the furnace jacket did not affect the specimens and rapid air cooling was achieved. The furnace element was wound noninductively in order to minimize the ac demagnetization effect of the heating current. The specimens were placed on silica shelves about 1 cm apart. Although the furnace could contain some 80 rock cylinders (19 × 19 mm), only 30 were used at a time, so that the variation in the field across the space occupied by the specimens was less than 5 γ. A detailed account of the technique has been given by *Chamalaun* [1963].

Results for the folded sites. As a first step a number of specimens from the folded sites, Manorbier Bay and Freshwater West, were thermally demagnetized. On the basis of the two-component hypothesis one would expect the inclination to decrease if the primary component was the thermally stable one or to increase if the secondary component was the more stable one. However, neither of these two things happened.

The specimens can be divided into two groups according to their thermal behavior. The first group, containing specimens whose direction of magnetization did not change significantly during thermal treatment, can be described as thermally stable specimens. The second group comprises specimens whose directions changed at higher temperatures. The important point to note is that the change in direction is consistent for specimens from the same hand sample. This can readily be seen from Figures 2, 3, and 4, which illustrate three further points of interest: (1) The change in directions is not that predicted by the two-component hypothesis, for in that case one would expect the directions to change toward a position close to $D = 210°$, $I = -4°$ (the mean direction of the flat lying sites) or to $D = 179°$, $I = +85°$ (the direction of the Permian field with respect to the bedding planes of the folded sites). (2) The directions above 550°C agree well with the directions obtained by *Stubbs* [1958] for the Old Red Sandstone lavas of Scotland. This is discussed below. (3) In the figures, the directions are plotted with respect to the bedding planes and it can be seen that there is much better agreement between specimens from the north and south sides of the fold after heating than before heating. Since the bedding planes are almost vertical the directions would be almost 180° apart if referred to the present horizontal. This application of the fold test [*Graham*, 1949] shows that the high-temperature directions were acquired before folding.

Results for the flat-lying sites. It seemed pertinent to investigate whether the flat-lying sites contained the same high-temperature component as the folded sites. Thermal demagnetization experiments were therefore conducted on specimens from 4 sites, Cwmbran, Portishead, Nant Ddu, and the Brecon Beacons. Specimens from the Brecon Beacons were heated in steps, but those from the other sites were heated directly to 600°C and cooled in zero field. Except for a number of specimens from the Brecon Beacons, most specimens were thermally unstable. The latter yielded the same high-temperature directions as found from the folded sites. This can be seen in Figures 5, 6, 7, and 8, in which the directions are plotted with respect to the bedding planes. The fact that the high-temperature directions for the folded and flat-lying sites agree when referred to the bedding planes is conclusive evidence that they were acquired before folding.

Comparison with Stubbs' results. The high-temperature directions are shown in Figure 9, together with a circle of confidence at the 95 per cent probability level [*Fisher*, 1953]. The

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Fig. 3. Change in direction for specimens from sample FW1 (Freshwater West).

Fig. 4. Change in direction for specimens from sample FW2.

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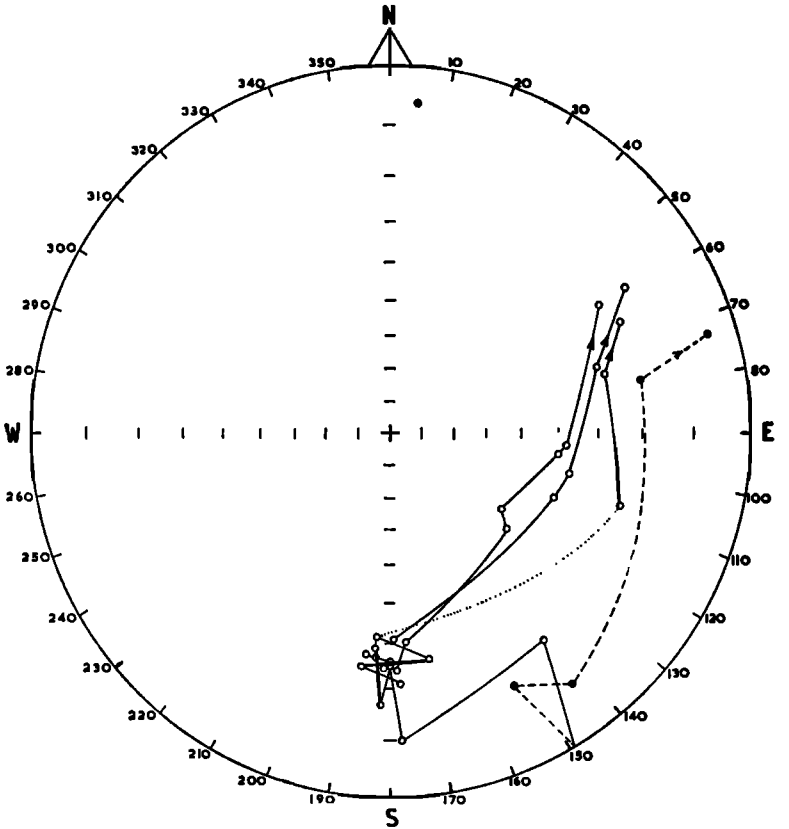


Fig. 5. Change in direction for specimens from sample BR26 (Brecon Beacons).

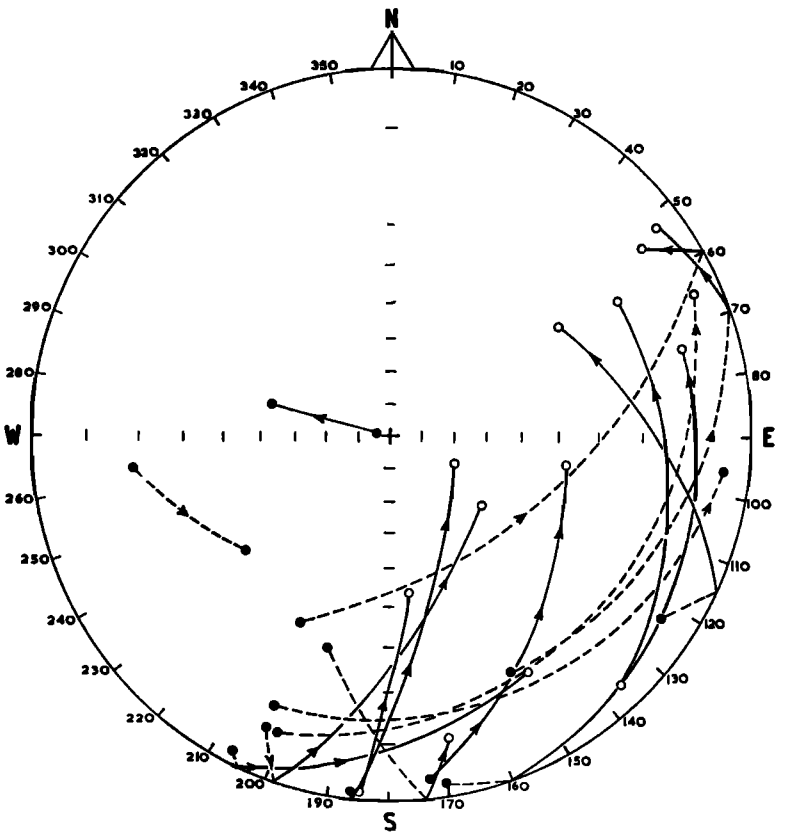


Fig. 6. Change in direction for specimens of Cwmbran.

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Fig. 8. Change in direction for specimens of Portishead.

Fig. 7. Change in direction for specimens of Nant Du.

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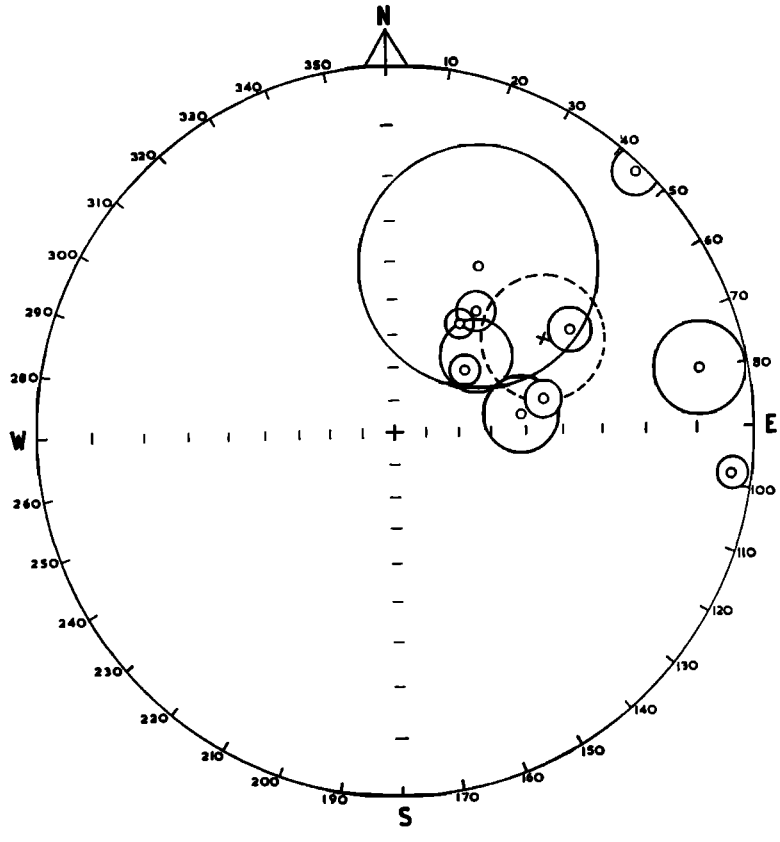


Fig. 9. Directions of the Lower Old Red Sandstone lavas. Mean direction $D = 60°$, $I = 38°$ (indicated by cross). Dotted circle indicates circle of confidence at 95 per cent probability level of the mean. Solid circles indicate circle of confidence at 50 per cent probability level for each flow.

scatter of directions results from several factors [*Chamalaun*, 1963]. First, the intensities have become weak, thus decreasing the accuracy with which the directions can be determined. Second, the susceptibility has increased (by a factor of 5 to 10), and this causes induced moments to become troublesome. Third, the fact that the direction continues to change on heating from 600°C to 650°C, indicates that the unstable component is not completely removed even at these temperatures. Fourth, when a temperature of 500°C is reached, the specimens become capable of acquiring a large component of isothermal remanence during storage. This was minimized by keeping the specimens in zero field almost until they were measured, but unfortunately they had to be transported over a distance of 24 km to the magnetometer. Therefore, and because the samples which show this high-temperature component are numerically few and do not constitute a sample of adequate size from this formation, we do not consider ourselves justified in calculating an alternative pole position from them. The fold test shows that the high-temperature direction was acquired before folding. Since the folding is Permo-Carboniferous in age this would indicate that the high-temperature component is either Carboniferous or Devonian in age.

It was pointed out above that the high-temperature directions agreed with the directions obtained by Stubbs for the Lower Old Red Sandstone lavas from Scotland. To illustrate this more clearly some of Stubbs' data are reproduced in Figure 10. For convenience only

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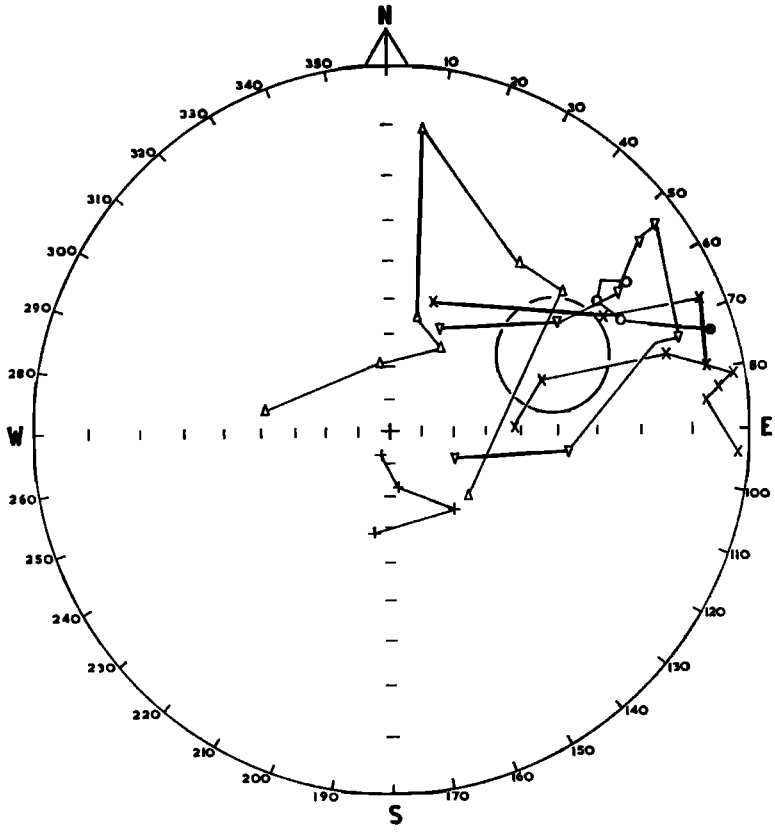


Fig. 10. High-temperature directions. Mean $D = 66°$, $I = 37°$. Upright triangles M5, inverted triangle Cwmbran, Cross Portishead, X's FW1 and FW2, circles BR26. Specimens from the same sample or site are connected by solid lines.

those directions with upward inclinations are shown, and it can be seen that the two mean directions are in excellent agreement.

Unless the magnetization of the Lower Old Red Sandstone lavas is actually a Carboniferous magnetization (for which there is no evidence), we believe that the combination of the fold test and igneous control constitutes strong evidence that both the high-temperature directions and Stubbs' data do in fact represent the Devonian field for Britain.

Discussion and conclusions. We interpret the results as indicating that a strong component of secondary magnetization has been imposed on the original Devonian magnetization of the Old Red Sandstone. The latter is probably indicated by the high-temperature directions. The secondary magnetization has completely obliterated the original magnetization in most samples. In a few samples a small and thermally stable part has been retained. It is therefore concluded that the Devonian paleomagnetic pole A is actually based on secondary magnetization and we suggest that pole B is the true Devonian paleomagnetic pole. The reliability of the latter is indicated by the mutual consistency between the high-temperature directions of the Old Red Sandstone sediments and the directions of the Lower Old Red Sandstone lavas of Scotland from which pole B was computed. Thus the discrepancy between the two apparently equally acceptable Devonian paleomagnetic poles for Britain is resolved.

The results reported here illustrate that thermal demagnetization is a powerful tool for discovering small but stable components of magnetization which are entirely masked in the NRM by an additional component of magnetization. However, for the method to be successful it is necessary that the heatings be

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continued up to the Curie temperature of the magnetic minerals. Also they have to be carried out on a large number of specimens, for only a few may actually have retained a memory of the original magnetization.

We suspect that there may be more red bed formations which have had a similar magnetic history to the Old Red Sandstones. Those formations, which are flat lying and on which the fold test cannot be performed, should especially need to be carefully re-examined, in particular since it was found here that the flat-lying sites had become remagnetized as well. No discussion has been given here of the mechanism of remagnetization and the implications of the results on the geometry of the magnetic vectors of the folded sites. These questions were examined by *Chamalaun* [1963] and will be the subject of a separate paper now in the course of preparation.

Finally, accepting the revised Devonian pole poses the new problem of the apparent agreement between Devonian data from other parts of Europe (notably Russia) and the secondary magnetization of the Old Red Sandstones of the Anglo-Welsh cuvette and the disagreement between these European data and the revised Devonian pole. The formulation of a satisfactory solution to this problem has not yet been possible because of too few data. In particular it would be interesting to see the age of the magnetization of the Russian rocks to be tested by means of the fold test.

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